

Correlation Functions in a Finite-Size Kitaev Model

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Abstract

In this note, we discuss how to compute multi-spin correlation functions in a finite-size Kitaev model, which only contains four distinct plaquettes. These correlation functions are crucial in constructing the density matrix of the Kitaev ground state.

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1 Foundations

1.1 Correlation Functions

As mentioned in our work [1], at the ground state, the multi-spin correlation function of the Kitaev model is finite if and only if it can be written in the form of a string. Therefore, we only need to discuss the following form of the correlation function

$$\left\langle \prod_{\langle ij \rangle \in \mathcal{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle$$

This kind of correlation function can be exactly evaluated in the Majorana representation. Substituting the transformation

$$\sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \rightarrow i b_i^{\alpha_{ij}} c_i i b_j^{\alpha_{ij}} c_j = \hat{u}_{ij} i c_j c_i$$

into the correlation function and taking the trivial gauge, we have

$$\left\langle \prod_{\langle ij \rangle \in \mathcal{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle = \left\langle \prod_{\langle ij \rangle \in \mathcal{D}} (-i) c_i c_j \right\rangle_M$$

1.2 Minimal Size for Non-trivial Topology

For numerical efficiency, we tend to choose the smallest possible system. However, the system must still be sufficiently large such that the topological order still exists. A requirement for this is that the ground state manifold should possess four independent topological sectors generated by global Wilson loops if the geometry of the system is a torus.

We start from the Wen plaquette model. There are two requirements for a Wen plaquette system to possess four topologically degenerate ground states:

- The lattice must contain an even number of rows or columns in both the horizontal and vertical directions.
- A global Wilson loop cannot be generated or destroyed by local plaquette operators.

The physical meaning of the first requirement is to preserve the checkerboard pattern of the lattice. If there are adjacent plaquettes (cells) that are of the same type (black or white, in a checkerboard sense), then the two types of plaquettes become identical, and the original two different Wilson loops become the same. This breakdown of the checkerboard pattern reduces the original fourfold topological ground state degeneracy down to twofold, which is not adequate to describe the physics of a toric code system.

According to the two requirements, **the smallest possible topologically non-trivial Wen plaquette model contains eight ($8 = 2 \times 4$) distinct plaquettes and eight sites**. A 2×2 checkerboard is not adequate for the system to have four topological sectors since it violates the second requirement¹.

¹The *global Wilson loop* is identical to a plaquette operator

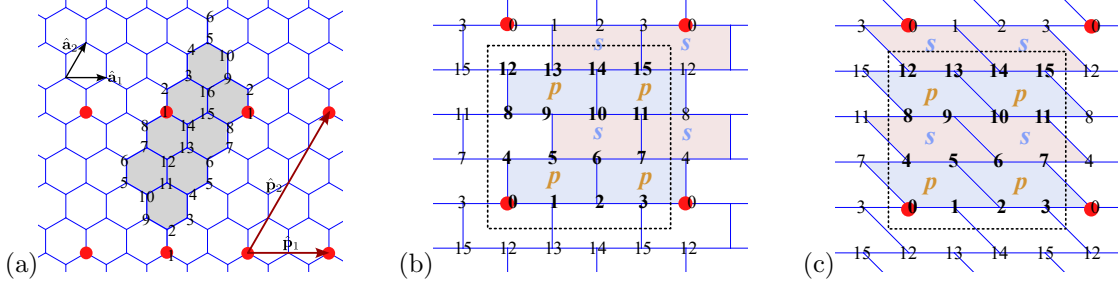


Figure 1: (a) The smallest topologically non-trivial system consists of 8 unit cells, which is equivalent to 16 spins. The two vectors that define the periodic boundary condition are $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, respectively. (b) For numerical simplicity, the honeycomb lattice is rearranged into a brick-wall geometry, and the spins are relabeled according to their Euclidean coordinates in the lattice. In both figures, red dots indicate identical lattice sites. (c) An equivalent choice.

2 Finite-Size Correlation

We only have to focus on the two-point correlation functions. Here we set that the system satisfies the **periodic boundary condition**, which is characterized by two independent vectors, $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, as illustrated in Fig. 1.a. The geometry adopted here follows the one illustrated in Ref. [2]

2.1 Momentum States in Finite Systems

For slave fermions in the parton construction, the boundary condition of the system could be periodic (PBC) or anti-periodic (APBC), depending on which topological sector we choose [3]. Here, we only consider the PBC in both the horizontal and vertical directions.

$$f_{\sigma\mu\mathbf{x}} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_1)} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_2)} \quad (1)$$

To determine the allowed momentum states, we first Fourier transform the fermions as follows

$$f_{\sigma\mu\mathbf{x}} = \frac{1}{\sqrt{N/2}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} f_{\sigma\mu\mathbf{k}}$$

With this expression, Eq. (1) implies that the allowed momentum $\mathbf{k}$ must satisfy

$$\begin{cases} \mathbf{k} \cdot \hat{\mathbf{p}}_1 = 2n_1\pi & ; \quad n_1 = 0, 1, \dots, N_h - 1 \\ \mathbf{k} \cdot \hat{\mathbf{p}}_2 = 2n_2\pi & ; \quad n_2 = 0, 1, \dots, N_v - 1 \end{cases} \quad (2)$$

In our case illustrated in Fig. 1, $N_h = 2$ and $N_v = 4$, representing the number of unit cells included in the system in the horizontal and vertical directions, respectively. Consider the unit vectors $\hat{\mathbf{a}}_1 = \hat{\mathbf{e}}_x$ and $\hat{\mathbf{a}}_2 = \frac{1}{2}\hat{\mathbf{e}}_x + \frac{\sqrt{3}}{2}\hat{\mathbf{e}}_y$. We can define the unit vectors in the momentum space as

$$\begin{cases} \hat{\mathbf{b}}_1 = 2\pi\hat{\mathbf{e}}_x - \frac{2\pi}{\sqrt{3}}\hat{\mathbf{e}}_y \\ \hat{\mathbf{b}}_2 = \frac{4\pi}{\sqrt{3}}\hat{\mathbf{e}}_y \end{cases}$$

such that $\hat{\mathbf{a}}_i \cdot \hat{\mathbf{b}}_j = 2\pi\delta_{ij}$. With these basis vectors, the allowed momentum states are

$$\mathbf{k} = \frac{n_1}{2}\hat{\mathbf{b}}_1 + \frac{n_2}{4}\hat{\mathbf{b}}_2 \quad ; \quad n_1 = 0, 1 \quad , \quad n_2 = 0, 1, 2, 3 \quad (3)$$

In general, if the relation between two representations $\mathbf{k} = k_x \hat{\mathbf{e}}_x + k_y \hat{\mathbf{e}}_y = k_1 \hat{\mathbf{b}}_1 + k_2 \hat{\mathbf{b}}_2$ is

$$\begin{cases} k_x = 2\pi k_1 \\ k_y = \frac{2\pi}{\sqrt{3}} (-k_1 + 2k_2) \end{cases}, \quad \begin{cases} k_1 = \frac{1}{2\pi} k_x \\ k_2 = \frac{1}{4\pi} (k_x + \sqrt{3}k_y) \end{cases}$$

2.2 Two-Points Correlation Functions

The correlation functions can be calculated more easily in the normal fermion language. As defined in the appendix, the c -Majorana fermions can be written in terms of γ -fermions,

$$\begin{cases} c_{\mathbf{x}A} = \gamma_{\mathbf{x}}^\dagger + \gamma_{\mathbf{x}} = \frac{1}{\sqrt{N/2}} \sum_{\mathbf{k}} \left(\gamma_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\mathbf{x}} + \gamma_{\mathbf{k}} e^{i\mathbf{k}\mathbf{x}} \right) \\ c_{\mathbf{x}B} = i(\gamma_{\mathbf{x}}^\dagger - \gamma_{\mathbf{x}}) = \frac{i}{\sqrt{N/2}} \sum_{\mathbf{k}} \left(\gamma_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\mathbf{x}} - \gamma_{\mathbf{k}} e^{i\mathbf{k}\mathbf{x}} \right) \end{cases}$$

And the ground state is

$$|\psi\rangle = A \exp \left(\sum_{k \in \text{HBZ}} g_k \gamma_k^\dagger \gamma_{-k}^\dagger \right)$$

where

$$A = \prod_{k \in \text{HBZ}} (1 + |g_k|^2)^{-1/2}$$

is the normalization constant. Similar to Nambu-Gor'kov Green's function, there are two types of fermion pairing, $AA(BB)$ and AB . Both types of correlation functions can be simplified to the problem of solving the γ -fermion correlation function as we will see later. The two γ -fermion correlation functions we have to know are $\langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle$ and $\langle \gamma_{\mathbf{k}}^\dagger \gamma_{k'} \rangle$. Their analytical forms are given as follows

$$\begin{cases} \langle \gamma_{\mathbf{k}}^\dagger \gamma_{k'} \rangle = \frac{|g_k|^2}{1 + |g_k|^2} \delta_{kk'} = v_k^2 \delta_{kk'} = \frac{1}{2} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \delta_{kk'} \\ \langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle = \frac{g_k}{1 + |g_k|^2} \delta_{k, -k'} = i \text{sgn}(I_k) u_k v_k \delta_{k, -k'} = i \frac{I_k}{2|s_{\mathbf{k}}|} \delta_{k, -k'} \end{cases}$$

and the fermionic commutation relations can give other two types of correlation functions

$$\begin{cases} \langle \gamma_{\mathbf{k}} \gamma_{k'}^\dagger \rangle = \delta_{kk'} - \langle \gamma_{k'}^\dagger \gamma_{\mathbf{k}} \rangle = u_k^2 \delta_{kk'} = \frac{1}{2} \left(1 + \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \delta_{kk'} \\ \langle \gamma_{\mathbf{k}}^\dagger \gamma_{k'}^\dagger \rangle = \langle \gamma_{k'} \gamma_{\mathbf{k}} \rangle^* = -i \text{sgn}(I_{k'}) u_{k'} v_{k'} \delta_{k', -k} = i \text{sgn}(I_k) u_k v_k \delta_{k, -k'} = \langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle \end{cases}$$

Here I list the explicit expression of s_k , R_k and I_k and other relevant quantities or functions. We start from

$$s_k = 2 (J_x e^{ik \cdot \hat{\mathbf{n}}_1} + J_y e^{ik \cdot \hat{\mathbf{n}}_2} + J_z)$$

Then, if we express the momentum state in terms of $\mathbf{k} = k_x \hat{\mathbf{e}}_x + k_y \hat{\mathbf{e}}_2$, then

$$\left. \begin{aligned} |s_k| &= 2 \left[J_x^2 + J_y^2 + J_z^2 + 2J_x J_y \cos(k_x) + 2J_x J_z \cos\left(\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) + 2J_y J_z \cos\left(\frac{1}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) \right]^{1/2} \\ R_k &= 2 \left[J_x \cos\left(\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) + J_y \cos\left(\frac{1}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) + J_z \right] \\ I_k &= 2 \left[J_x \sin\left(\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) - J_y \sin\left(\frac{1}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) \right] \end{aligned} \right\} \quad (4)$$

Or, expressed in terms of reciprocal basis vectors $\mathbf{k} = k_1 \hat{\mathbf{b}}_1 + k_2 \hat{\mathbf{b}}_2$, then

$$\left. \begin{aligned} |s_k| &= 2 [J_x^2 + J_y^2 + J_z^2 + 2J_x J_y \cos(2\pi k_1) + 2J_x J_z \cos(2\pi k_2) + 2J_y J_z \cos(2\pi(k_1 - k_2))]^{1/2} \\ R_k &= 2 [J_x \cos(2\pi k_2) + J_y \cos(2\pi(k_1 - k_2)) + J_z] \\ I_k &= 2 [J_x \sin(2\pi k_2) - J_y \sin(2\pi(k_1 - k_2))] \end{aligned} \right\} \quad (5)$$

Again, these results are general and are valid even in a finite-size system.

With these relations, we can start to compute the c -Majorana correlation functions. For AA -pairings, the correlation function is,

$$\begin{aligned} \langle c_{\mathbf{x}A} c_{\mathbf{x}'A} \rangle &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(\langle \gamma_{\mathbf{k}} \gamma_{-\mathbf{k}} \rangle + \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{\mathbf{k}}^\dagger \rangle + \langle \gamma_{\mathbf{k}} \gamma_{\mathbf{k}}^\dagger \rangle + \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{-\mathbf{k}} \rangle \right) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \\ &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} \\ &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \cos(\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})) \in \mathbb{R} \end{aligned}$$

The imaginary part is an odd function in the momentum space, hence it vanishes after the integration. However, this implies that the Majorana correlation function is a real number, which is impossible since

$$\langle c_i c_j \rangle^* = \langle c_j c_i \rangle = -\langle c_i c_j \rangle$$

which must be an imaginary number unless $i = j$. So we conclude that

$$\langle c_{\mathbf{x}A} c_{\mathbf{x}'A} \rangle = \langle c_{\mathbf{x}B} c_{\mathbf{x}'B} \rangle = \delta_{\mathbf{x}\mathbf{x}'} \quad (6)$$

As for AB -pairings, a similar calculation gives

$$\begin{aligned} \langle c_{\mathbf{x}A} c_{\mathbf{x}'B} \rangle &= \frac{i}{N/2} \sum_{\mathbf{k}} \left(\langle -\gamma_{\mathbf{k}} \gamma_{-\mathbf{k}} \rangle + \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{\mathbf{k}}^\dagger \rangle + \langle \gamma_{\mathbf{k}} \gamma_{\mathbf{k}}^\dagger \rangle - \langle \gamma_{-\mathbf{k}}^\dagger \gamma_{-\mathbf{k}} \rangle \right) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \\ &= \frac{i}{N/2} \sum_{\mathbf{k}} \left(\frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} - i \frac{I_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} = \frac{i}{N/2} \sum_{\mathbf{k}} \frac{s_{\mathbf{k}}^*}{|s_{\mathbf{k}}|} e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} \end{aligned}$$

This result is purely imaginary, and can be further simplified,

$$\langle -i c_{\mathbf{x}A} c_{\mathbf{x}'B} \rangle = \frac{1}{N/2} \sum_{\mathbf{k}} \cos[\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x}) - \theta_{\mathbf{k}}] = \frac{1}{N/2} \sum_{\mathbf{k}} \frac{1}{|s_{\mathbf{k}}|} [R_{\mathbf{k}} \cos(\mathbf{k} \cdot \delta \mathbf{x}) + I_{\mathbf{k}} \sin(\mathbf{k} \cdot \delta \mathbf{x})] \quad (7)$$

where $s_{\mathbf{k}} = |s_{\mathbf{k}}| e^{i\theta_{\mathbf{k}}}$. Notice that the above formulation is general and can be used directly in the case of finite systems.

2.3 Two-Points Correlation Functions with Magnetic Field

In the isotropic limit, applying an external magnetic field makes the Kitaev model equivalent to a Haldane honeycomb model [4] in the Majorana fermion representation. In general, the correlation functions of AA and BB pairings are finite **continue**. However, since the $\mathbb{Z}_2$ gauge fields are still c-numbers, the ground state still exhibits the pattern that only string-like operators bear finite expectation values. The only difference is that, in the situation with a magnetic field, the N.N.N. coupling contributes to the correlation functions of longer strings, and strings with both endpoints of the same type have finite expectation values in general.

2.4 Two-Points Correlation Functions in a Finite-Size System

In the following discussions, we consider the lattice geometry illustrated in Fig. 1.c. In this situation, there are three methods for labeling the position of lattice sites.

- The first method is to assign a number to each lattice site one by one. The label is denoted by n in this note.
- In the lattice geometry illustrated in Fig. 1.c, the lattice sites can be sorted into layers labeled by l , and the horizontal position of a lattice site in each layer is labeled by s , standing for *site*. For example, the position of site number 7 is $(l, s) = (1, 3)$, and the position of site number 12 is $(l, s) = (3, 0)$. The integer s can be further decomposed as

$$s = 2s_q + s_r$$

where $s_q \in \mathbb{N}$ and $s_r = 0 \vee 1$ is the remainder.

- With the Bravais lattice site expressed as $\mathbf{x} = x\hat{\mathbf{a}}_1 + y\hat{\mathbf{a}}_2$, each lattice site corresponds to a unique pair $(\mathbf{x}, \mu)$, where $\mu = 0, 1$, with 0 representing sublattice B and 1 representing A , respectively.
- The following equations describe the relationships among the three labeling methods listed above.

$$\begin{cases} n = s + lN_h \\ \mu = n \mod 2 \\ x = s_q + s_r \\ y = l - s_r \end{cases}$$

With this definition, the factor $\mathbf{k} \cdot \delta\mathbf{x}$ in Eq. (7) can be written as

$$\mathbf{k} \cdot \delta\mathbf{x} = \pi \left(n_1 \delta x + \frac{n_2}{2} \delta y \right)$$

where n_1 and n_2 are defined by the expression Eq. (3) of momentum states, and δx and δy are components of the position $\mathbf{x}$ represented in terms of lattice unit vectors $\mathbf{x} = \delta x \hat{\mathbf{a}}_1 + \delta y \hat{\mathbf{a}}_2$. Using this representation, the summation $\sum_{\mathbf{k}}$ in Eq. (7) can be written as $\sum_{n_1, n_2}$. The only difference is that the momentum is now represented in terms of integers n_1 and n_2 .

3 String Operators

3.1 Constructing String Operators

To define string operators, we have to define the **order** of multiplying the assembly of $\mathbb{Z}_2$ dimer operators to make each string operator uniquely defined.

Given the fact that a class $[\mathfrak{D}] \in \mathbf{G}$ in the **good set** $\mathbf{G}$, in which $\text{EP}(\mathfrak{D})$ contains the same amounts of the type- A and type- B endpoints,

$$\prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} \left(\prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right)$$

Algorithm 1 Sorting lattice sites into A and B sublattices

```

function SUBLATID( $n$ )           ▷ Define a function that identifies in which sublattice  $\mu$  the site  $n \in \mathcal{N}$  is.
    return  $\mu$ 
end function
 $\text{EP}_A(\mathfrak{D}) \leftarrow \{\}; \text{EP}_B(\mathfrak{D}) \leftarrow \{\}$            ▷ The sets that contains type- $A$  and type- $B$  sites, respectively
for  $n \in \mathfrak{D}$  do
    if SUBLATID( $n$ ) =  $A$  then
         $\text{EP}_A(\mathfrak{D}) \leftarrow \text{EP}_A(\mathfrak{D}) \cup \{n\}$ 
    else if SUBLATID( $n$ ) =  $B$  then
         $\text{EP}_B(\mathfrak{D}) \leftarrow \text{EP}_B(\mathfrak{D}) \cup \{n\}$ 
    end if
end for

```

Algorithm 2 Generate $\text{St}(\mathfrak{D})$

```

 $\text{St}(\mathfrak{D}) \leftarrow \{\}$ 
for  $i = 0, i \leq |\text{EP}(\mathfrak{D})|/2, i++$  do
     $n_a \leftarrow \text{EP}_A(\mathfrak{D})[i]; n_b \leftarrow \text{EP}_B(\mathfrak{D})[i]$ 
     $\text{St}(\mathfrak{D}) \leftarrow \text{St}(\mathfrak{D}) \cup \{(n_a, n_b)\}$ 
end for

```

Algorithm 3 Dimer operators

Require: n and n' must be nearest neighbor, and $\mu(n) = A$ by default

```

 $(l, s) \leftarrow \text{Dic}(n); (l', s') \leftarrow \text{Dic}(n')$ 
if  $l' = l - 1$  then
     $\text{dimer}(n, n') = \sigma_n^z \sigma_{n'}^z$ 
else if  $s' = s + 1$  then
     $\text{dimer}(n, n') = \sigma_n^x \sigma_{n'}^x$ 
else if  $s' = s - 1$  then
     $\text{dimer}(n, n') = \sigma_n^y \sigma_{n'}^y$ 
end if

```

Algorithm 4 Generate the string operator of a single string

```
(la, sa) ← Dic(na); (lb, sb) ← Dic(nb)           ▷ The function Dic translates a lattice label n into the layer-site representation
Σ∅ ← 1                                           ▷ Initialize the string operator
(l, s) ← (la, sa)                               ▷ Initialize the moving point
while (l, s) ≠ (lb, sb) do
  if lb = la then
    if sb > sa then
      (l', s') ← (l, s + 1)
    else if sb < sa then
      (l', s') ← (l, s - 1)
    end if
  else if lb > la then           ▷ Move the moving point to the layer lb first, and then shift it to the right site sb.
    if the current moving point is in B then
      (l', s') ← (l + 1, s - 1)
    else if the current moving point is in A then
      (l', s') ← (l, s + 1)
    end if
  else if lb < la then           ▷ Move the moving point to the layer lb first, and then shift it to the right site sb.
    if the current moving point is in B then
      (l', s') ← (l - 1, s + 1)
    else if the current moving point is in A then
      (l', s') ← (l, s + 1)
    end if
  end if
  n ← Dic-1(l, s); n' ← Dic-1(l', s')
  Σ∅ ← Σ∅ · dimer(n, n')
  (l, s) ← (l', s')
end while
```

Notice that the periodic boundary condition requires

$$s + N_h = s, \quad l + N_v = l$$

Therefore, the function Dic^{-1} must satisfy

$$\text{Dic}^{-1}(s + N_h, l) = \text{Dic}^{-1}(s, l + N_v) = \text{Dic}^{-1}(s, l)$$

3.2 String Correlation Functions

$$\prod_{\langle ij \rangle \in \mathcal{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C} \in \text{St}(\mathcal{D})} \left(\prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right)$$

We have required $\mathcal{C} \in \mathbf{G}$. That is to say, each string $\mathcal{C}$ connects a type-*A* and a type-*B* endpoint.

$$\begin{aligned} \prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} &\rightarrow \prod_{\langle ij \rangle \in \mathcal{C}} (-i) c_i c_j = (-i)^{|\mathcal{C}|} (-1)^{(|\mathcal{C}|-1)/2} c_{a_{\mathcal{C}}} c_{b_{\mathcal{C}}} \\ &= -i (-1)^{|\mathcal{C}|(|\mathcal{C}|-1)/2} (-1)^{(|\mathcal{C}|-1)/2} c_{a_{\mathcal{C}}} c_{b_{\mathcal{C}}} \\ &= -i (-1)^{(|\mathcal{C}|+1)^2/2} c_{a_{\mathcal{C}}} c_{b_{\mathcal{C}}} \end{aligned}$$

In the derivation, we have used the identity $i^n = (-i)^n = (-1)^{n(n-1)/2}$ if n is an even integer. The sign factor $(-1)^{(|\mathcal{C}|-1)/2}$ is due to reversing the direction of dimers to make them point from *B* to *A*.

Notice that in each dimer $\sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}}$, the former index i represents the lattice site in the A -sublattice, while the latter one j stands for the B -sublattice, and the direction of the string $\mathcal{C}$ is always from its type- A endpoint to the type- B one. At the same time, because the two endpoints of each string are in different sublattices, considering the honeycomb lattice geometry, the length $|\mathcal{C}|$ of each string $\mathcal{C}$ must be an odd number. With this property, the factor $(|\mathcal{C}| + 1)^2/2$ that appears in the exponent of the sign factor must be a multiple of 4, making the sign $(-1)^{(|\mathcal{C}|+1)^2/2}$ trivial. Therefore, we eventually obtain the correspondence between a string operator and its Majorana representation, which depends only on the endpoints $a_{\mathcal{C}}$ and $b_{\mathcal{C}}$ if $\mathcal{C}$ is a *good* string

$$\prod_{\langle ij \rangle \in \mathcal{C} \subset \mathbf{G}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \rightarrow -i c_{a_{\mathcal{C}}} c_{b_{\mathcal{C}}} \quad (8)$$

$$\prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} \left(\prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right) \rightarrow \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} (-i c_{a_{\mathcal{C}}} c_{b_{\mathcal{C}}}) \quad (9)$$

$$\begin{aligned} \mathfrak{C}_{\mathfrak{D}} &= \left\langle \prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle \rightarrow \left\langle \prod_{\mathcal{C} \in \text{St}(\mathfrak{D})} (-i c_{a_{\mathcal{C}}} c_{b_{\mathcal{C}}}) \right\rangle_M \\ &= \begin{vmatrix} \langle -i c_{a_1} c_{b_1} \rangle_M & \langle -i c_{a_1} c_{b_2} \rangle_M & \langle -i c_{a_1} c_{b_3} \rangle_M & \cdots \\ \langle -i c_{a_2} c_{b_1} \rangle_M & \langle -i c_{a_2} c_{b_2} \rangle_M & \langle -i c_{a_2} c_{b_3} \rangle_M & \cdots \\ \langle -i c_{a_3} c_{b_1} \rangle_M & \langle -i c_{a_3} c_{b_2} \rangle_M & \langle -i c_{a_3} c_{b_3} \rangle_M & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} \end{aligned} \quad (10)$$

4 Genuine Multipartite Negativity as an Entanglement Monotone

Definition 4.1 (Separability [5]). A state ϱ is called **separable** if there exists a bipartition of the Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$ such that

$$\varrho = \sum_n p_n \rho_n^{(A)} \otimes \rho_n^{(\bar{A})} \quad (11)$$

where each component $\rho_n^{(A/\bar{A})}$ is a density matrix acting on the subspace $\mathcal{H}_A$ and $\mathcal{H}_{\bar{A}}$. Weights p_n are convex, satisfying $0 \leq p_n \leq 1 \forall n$ and $\sum_n p_n = 1$. In other words, $\{p_n\}$ is a *probability distribution*. A state that is separable with respect to a bipartition $A|\bar{A}$ is denoted by $\varrho_{A|\bar{A}}^{\text{sep}}$. Such a state is often referred to as a *classically correlated* state.

Here, I adopt the definition used in Ref. [5]. The components $\rho_n^{(A/\bar{A})}$ may or may not be pure states. In this aspect, this definition is different from that used in Ref. [6] and Ref. [7], where the definition of separability requires those components to be pure. Later, in Lemma 4.3, we will see that **the separable states given by Eq. (11) are states that have zero negativity**.

Definition 4.2 (Weak Biseparability [6]). A state ϱ is called **weakly biseparable** if it can be expressed as a mixture of separable states defined by Def. 4.1

$$\varrho = \sum_{A \subset \Lambda} p_A \varrho_{A|\bar{A}}^{\text{sep}} \quad (12)$$

where $\{p_A\}$ is a probability distribution.

Definition 4.3 (Strong Biseparability [7]). Ref. [7] gives a seemingly different definition. A state ϱ is called **strongly biseparable** if it can be expressed as a mixture of **biseparable pure states**

$$\varrho = \sum_{A \subset \Lambda} p_A |\psi\rangle_A \langle\psi|_A \otimes |\phi\rangle_{\bar{A}} \langle\phi|_{\bar{A}} \quad (13)$$

where $\{p_A\}$ is a probability distribution.

Compared to Def. 4.2, Def. 4.3 covers a narrower range since separable states $\varrho_{A|\bar{A}}^{(\text{sep})}$ in Eq. (12) are not required to be pure. Later, we will see that genuine multipartite negativity, as a monotone of multipartite entanglement, vanishes if a state is biseparable defined by Def. 4.2. In other words, **the condition required by Def. 4.3 is too strong**. The range of states that are not genuinely multipartite entangled is wider than the range restricted by Def. 4.3.

Lemma 4.1. Suppose a system Λ can be divided into n parties $\Lambda \rightarrow A_1|A_2|\cdots|A_n$, such a division is denoted by $\mathcal{D}_n$, and there exists a series of density matrices $\{\rho_k^{(A_1)}, \rho_k^{(A_2)}, \dots, \rho_k^{(A_n)}\}$ and a probability distribution $\{p_k\}$ such that the following convex mixture

$$\varrho = \sum_k p_k \bigotimes_{A \in \mathcal{D}_n} \rho_k^{(A)} \quad (14)$$

is also a density matrix, then ϱ defined in the above equation is *weakly biseparable*.

Proof. The proof is trivial since each component $\bigotimes_{A \in \mathcal{D}_n} \rho_k^{(A)}$ in Eq. (14) is separable. ■

Definition 4.4 (Genuine Multipartite Entanglement). A state is **genuinly multipartite entangled** if it is **not biseparable**. Note that the definition of *biseparable* adopted here is based on Def. 4.2.

Recently, Ref. [7] introduces a concept of **entanglement frustration** in **quantum spin liquid** phases, which states that genuine multipartite entanglement can be found only in **loopy** regions that contain closed loops, contrasting the cases in **non-loopy regions**, where genuine multipartite entanglement is significantly suppressed, which may be understood that the entanglement is *frustrated* by the quantum fluctuation. To describe this phenomenon, the authors of Ref. [7] adopt **genuin multipartite negativity** to quantitatively measure the multipartite entanglement.

Definition 4.5 (Negativity). For bipartite entanglement, the **negativity** of a state ρ is defined as [8, 9]

$$N_A(\varrho) \equiv \frac{\|\varrho^{TA}\|_1 - 1}{2}$$

where $\|A\|_1 \equiv \text{tr}(\sqrt{A^\dagger A})$ is the trace form of a matrix.

Lemma 4.2. The negativity of a state is always positive, and the value is equal to the sum of the absolute value of negative eigenvalues of its partial transpose.

Proof. First, separate the eigenvalues of ϱ^{TA} into $R_+ \cup R_-$, where R_+ contains only positive eigenvalues and R_- contains only non-positive ones. By definition, the negativity can be expressed in terms of the eigenvalues of ϱ^{TA}

$$N_A(\varrho) = \frac{1}{2} \left(\sum_{r \in R_+ \cup R_-} \sqrt{r^2} - 1 \right) = \frac{1}{2} \left(\sum_{r_+ \in R_+} r_+ - \sum_{r_- \in R_-} r_- - 1 \right) = \sum_{r_- \in R_-} |r_-| \geq 0$$

where we have used the facts that a state traces to one and a partial transpose is a trace-preserving operation. ■

Definition 4.6. A *minimum negativity* is the smallest possible negativity among all possible bipartitions of the system $\Lambda \rightarrow A|\bar{A}$

$$N^{\min}(\varrho) \equiv \min_{A \subset \Lambda} N_A(\varrho)$$

Such a quantity does not depend on the partition. If the Hilbert space $\mathcal{H}$ is restricted in such a way that it can only be separated into n **local subspaces (parties)** $\mathcal{H} = \bigotimes_{k=1}^n \mathcal{H}_k$, then there are $2^{n-1} - 1$ different ways to bipartition the system. If such a constraint is provided, then the subregion A is required to be a composition of the given elementary regions. In this case, the minimum negativity is specifically called an *n -party minimum negativity*, denoted by $N_n^{\min}(\varrho)$.

The minimum negativity encodes information about the genuine multipartite entanglement, as it searches over all possible system separations. However, it is not adequate as a monotone that quantifies the GME since $N^{\min}(\varrho) > 0$ does not guarantee GME. To achieve this purpose, the authors of [6] and [7] came up with the following quantity that modifies the previous definition.

Lemma 4.3. A separable state has zero minimum negativity.

Proof. According to Def. 4.1, if a state ϱ is separable, then there exists at least a bipartition $\Lambda \rightarrow A|\bar{A}$ such that ϱ can be expressed in the form of Eq. (11). Take such a bipartition to compute the negativity $N_A(\varrho)$, we obtain

$$N_A(\varrho) = \frac{1}{2} \left(\left\| \sum_n p_n (\rho_n^{(A)})^{T_A} \otimes \rho_n^{(\bar{A})} \right\|_1 - 1 \right)$$

Because $\rho_n^{(A)}$ is a density matrix, its transpose is still a density matrix $(\rho_n^{(A)})^{T_A} = \rho_n^{(A)*}$. With this property, the density matrix after the partial transpose is still a valid density matrix satisfying $\text{Tr}(\varrho^{T_A}) = 1$ and with all the eigenvalues non-negative. According to Lemma 4.2, the resulting negativity is zero. Because negativity is always non-negative, this result implies that $N_A(\varrho)$ reaches its minimum value when considering the bipartition $A|\bar{A}$. ■

Definition 4.7 (Genuine Multipartite Negativity (GMN) [7]). Denote by $\mathcal{DC} \equiv \{\{\rho_k\}\}$ the set of *decomposition* $\{\rho_k\}$ of the state ϱ , where each decomposition can expand the density matrix as $\varrho = \sum_k p_k \rho_k$ with $\{p_k\}$ a probability distribution, then the *genuine multipartite negativity* (GMN) of a state ϱ , denoted by $\mathcal{N}(\varrho)$, is defined as

$$\mathcal{N}(\varrho) \equiv \min_{\{\rho_k\} \in \mathcal{DC}} \sum_k p_k N^{\min}(\rho_k) \quad (15)$$

A *n -party genuine multipartite negativity*, denoted by $\mathcal{N}_n(\varrho)$, is defined in the same way as Eq. (15), with each $N^{\min}(\rho_k)$ being replaced with the n -party genuine multipartite negativity $N_n^{\min}(\rho_k)$.

Lemma 4.4. A biseparable state has zero GMN. Note that the definition of biseparability adopted here is based on Def. 4.2.

Proof. If a state ρ is biseparable, then it can be expressed in the form given by Eq. 12. Evaluating the mixture of the negativity of the separable components in Eq. (12)

$$\sum_k p_k N^{\min} \left(\rho_{A_k|A_k}^{(\text{sep})} \right) = 0$$

gives zero because, according to Lemma 4.3, the minimum negativity of a separable state is zero. Because the lower limit of the negativity is zero, the previous observation implies that the biseparable decomposition Eq. (12) gives the minimum in Eq. (15) directly. ■

Theorem 4.5 (GMN as a Monotone). *A state is genuinely multipartite entangled if its GMN is finite.*

Proof. Lemma. 4.4 implies that a state having a finite GMN must not be biseparable. According to Def. 4.4, a genuinely multipartite state is a state that is not biseparable. Combining these two facts proves this corollary. ■

The above definition can be found in Ref. [7]. The searching of the separation $\{\rho_k\}$ that minimizes the functional $\sum_k p_k N^{\min}(\rho_k)$ can be implemented using **semidefinite programming** introduced in Ref. [6, 10].

Definition 4.8 (Fully Decomposable Witness). *A entanglement witness is fully decomposable if $\forall A \subset \Lambda$, there exist **positive semidefinite** [11] operators P and Q such that*

$$W = P + Q^{T_A}$$

where T_A stands for the partial transpose with respect to A . Ref. [7] adds one more constraint to the operator Q that $Q \leq \mathbb{1}$.

The definition of fully decomposable witnesses can be used to reformulate the genuine multipartite negativity that can be computed numerically more easily.

Definition 4.9 (Genuine Multipartite Negativity [6, 10]). Given $\mathfrak{W}$ the set of all decomposable witnesses, the genuine multipartite entanglement of a state ρ is defined as

$$\mathcal{N}(\rho) \equiv - \min_{W \in \mathfrak{W}} \text{Tr}(\rho W)$$

Moreover, Ref. [7] further shows that the set $\mathfrak{W}$ can be reduced to a set that only contains the partial transpose, $\mathfrak{W} = \{Q_A^{T_A} \forall A : 0 \leq Q_A \leq \mathbb{1}\}$ of semidefinite operators [11].

Theorem 4.6. *Definitions 4.7 and 4.9 are equivalent.*

Proof. We start by discussing an expression equivalent to the minimum negativity defined in Def. 4.5. First, decompose the partial transpose of the density matrix $\rho^{T_A} = \mathcal{O}_+ + \mathcal{O}_-$ into a semidefinite $\mathcal{O}_+ \geq 0$ and a negative part $\mathcal{O}_- < 0$. By definition, the negativity $N(\rho)$ is equal to the absolute value of the trace of the negative part $|\text{Tr}(\mathcal{O}_-)| = -\text{Tr}(\mathcal{O}_-)$. Suppose an operator $0 \leq Q \leq \mathbb{1}$ is semidefinite, then the composition $\mathcal{O}_+Q$ is always semidefinite, while $\mathcal{O}_-Q$ is always negative, rendering the following expression of the trace

$$-\text{Tr}(\rho^{T_A}Q) = -\text{Tr}(\mathcal{O}_+Q) - \text{Tr}(\mathcal{O}_-Q) = -|\text{Tr}(\mathcal{O}_+Q)| + |\text{Tr}(\mathcal{O}_-Q)|$$

$$\begin{aligned}
-\min_{0 \leq Q \leq \mathbb{1}} \text{Tr}(\varrho^{T_A} Q) &= -\min_{0 \leq Q \leq \mathbb{1}} |\text{Tr}(\mathcal{O}_+ Q)| + \max_{0 \leq Q \leq \mathbb{1}} |\text{Tr}(\mathcal{O}_- Q)| \\
|\text{Tr}(\mathcal{O}_- Q)| &\leq \left| \text{Tr}(\mathcal{O}_-^\dagger \mathcal{O}_-) \right|^{1/2} |\text{Tr}(Q^\dagger Q)|^{1/2}
\end{aligned}$$

Because both $(-\mathcal{O}_-)$ and Q are positive semidefinite, we have

$$\left| \text{Tr}(\mathcal{O}_-^\dagger \mathcal{O}_-) \right|^{1/2} |\text{Tr}(Q^\dagger Q)|^{1/2} \leq |\text{Tr} \mathcal{O}_-| |\text{Tr} Q|$$

Altogether, we obtain

$$\max_{0 \leq Q \leq \mathbb{1}} |\text{Tr}(\mathcal{O}_- Q)| = |\text{Tr} \mathcal{O}_-| \max_{0 \leq Q \leq \mathbb{1}} |\text{Tr} Q| = |\text{Tr} \mathcal{O}_-| = N(\varrho)$$

$$-\min_{-\leq Q \leq \mathbb{1}} \text{Tr}(\varrho^{T_A} Q) = -\min_{-\leq Q \leq \mathbb{1}} \text{Tr}(\varrho Q^{T_A}) = N(\varrho)$$

continue

$$\begin{aligned}
\mathcal{L}(\{\rho_k\}, W) &= \sum_k p_k N^{\min}(\rho_k) + \text{Tr} \left(W \underbrace{\left(\rho - \sum_k p_k \rho_k \right)}_{\text{constraint}} \right) \\
&= \text{Tr}(W \rho) + \sum_k p_k (N^{\min}(\rho_k) - \text{Tr}(W \rho_k))
\end{aligned}$$

Because $p_k \geq 0 \forall k$, the dual problem requires

$$N^{\min}(\rho_k) - \text{Tr}(W \rho_k) \geq 0$$

continue

■

5 Gauge Fields and Loopy Entanglement Pattern

6 Pauli Stabilizer Projectors and GME

In Ref. [7], *entanglement frustration* consists of two major aspects:

1. In a quantum spin liquid phase, the GME is suppressed in non-loopy regions.
2. In a quantum spin liquid phase, the GME emerges in loopy regions.

Based on their observation, we would like to ask two questions:

1. Which feature of a quantum spin liquid phase causes the suppression of GME in non-loopy regions?
 - Symmetry-preserving ground state
 - Topological ground state degeneracy

- Fractionalization of excitations
 - Anyon statistics (which is also related to the ground state degeneracy)
2. For what reason do loops matter? Why does it stabilize the GME?

two conjectures

1. Symmetry-preserving density matrices that bear string-like forms are not genuinely multipartite entangled in regions that do not contain loops.
2. A stabilizer projector as a density matrix is genuinely multipartite entangled.

6.1 General Phenomena

Lemma 6.1. A density matrix of the following form

$$\rho = \frac{\mathbb{1}_\Lambda + \Sigma_\Lambda}{2^{|\Lambda|}} \quad (16)$$

is **not** genuinely multipartite entangled, where Σ_Λ is a Pauli string operator supported on the subsystem Λ .

Proof. The system Λ can be separated into the bipartitioned $A|\bar{A}$. In this way, the density matrix Eq. (16) can be rewritten in the following form

$$\rho = \frac{\mathbb{1}_A \mathbb{1}_{\bar{A}} + \Sigma_A \Sigma_{\bar{A}}}{2^{|\Lambda|+|\bar{\Lambda}|}}$$

where the two Pauli strings Σ_A and $\Sigma_{\bar{A}}$, supported on the subregion A and $\bar{A}$, respectively, combine to form the original Pauli string Σ_Λ supported on the initial system. In this expression, the density matrix can be separated directly

$$\rho = \frac{1}{2} \left(\frac{\mathbb{1}_A + \Sigma_A}{2^{|\Lambda|}} \right) \left(\frac{\mathbb{1}_{\bar{A}} + \Sigma_{\bar{A}}}{2^{|\bar{\Lambda}|}} \right) + \frac{1}{2} \left(\frac{\mathbb{1}_A - \Sigma_A}{2^{|\Lambda|}} \right) \left(\frac{\mathbb{1}_{\bar{A}} - \Sigma_{\bar{A}}}{2^{|\bar{\Lambda}|}} \right)$$

which is separable with respect to bipartition $A|\bar{A}$, according to Def. 4.1, and the size of the subsystem A is arbitrary. Since a separable state is by definition a weakly biseparable state, according to Def. 4.4, the above result completes the proof. ■

Corollary 6.1.1. The ground-state density matrix of a $\mathbb{Z}_2$ string net model (toric code model) in a subsystem containing only one plaquette or star is not genuinely multipartite entangled.

Lemma 6.1 and its corollary show that **a single gauge loop (Wilson loop) without matter in a $\mathbb{Z}_2$ gauge theory is not adequate to produce GME, regardless of how large the loop is.**

6.2 Biseparability in Kitaev Model

We start with a subsystem containing only two adjoint edges in an infinitely large Kitaev model. The density matrix, according to our work [1], reads as

$$\varrho = \frac{1}{2^3} (\mathbb{1} + \langle \sigma_1^x \sigma_2^x \rangle \sigma_1^x \sigma_2^x + \langle \sigma_3^y \sigma_2^y \rangle \sigma_3^y \sigma_2^y) =: \frac{1}{8} (\mathbb{1} - x \sigma_1^x \sigma_2^x + y \sigma_3^y \sigma_2^y) \quad (17)$$

$$\varrho_s = \sum_k p_k \bigotimes_{i \in \Lambda} \left(\frac{\mathbb{1} + \mathbf{v}_i^{(k)} \cdot \boldsymbol{\sigma}_i}{2} \right) = \frac{1}{2^{|\Lambda|}} \sum_{A \subset \Lambda} \sum_{\{\mu\}} \left(\sum_k p_k \prod_{i \in A} v_i^{(k)\mu_i} \right) \bigotimes_{i \in A} \sigma_i^{\mu_i}$$

$\mu=x,y,z$

A Exactly Solving for the Ground State

According to equation (29) of Ref. [12], we can diagonalize the Kitaev Hamiltonian in the momentum space if the gauge is fixed to be $u_{ij} = +1 \forall (ij)$,

$$\hat{H} = \sum_{\mathbf{k}} \Psi_k^\dagger U(k) \Psi_k = \frac{1}{2} \sum_{\mathbf{k}} \left[i s_{\mathbf{k}} d_{k,A}^\dagger d_{k,B} - i s_{\mathbf{k}}^* d_{k,B}^\dagger d_{k,A} \right]$$

where

$$\Psi_k \equiv \frac{1}{\sqrt{2N}} \sum_{\mathbf{r}} e^{-ik \cdot \mathbf{r}} \begin{pmatrix} c_{\mathbf{r},A} \\ c_{\mathbf{r},B} \end{pmatrix} = \begin{pmatrix} d_{k,A} \\ d_{k,B} \end{pmatrix}$$

$$U(k) = \frac{1}{2} \begin{pmatrix} 0 & i s_{\mathbf{k}} \\ -i s_{\mathbf{k}}^* & 0 \end{pmatrix}, \quad s_{\mathbf{k}} = 2 \left(J_x e^{ik \cdot (\mathbf{a}_2 - \mathbf{a}_1)} + J_y e^{ik \cdot \mathbf{a}_2} + J_z \right)$$

Note that the index $\mathbf{r}$ above is the Bravais lattice point, and A, B represent the sublattice. If we further define a normal fermion operator,

$$\gamma_{\mathbf{r}} \equiv \frac{1}{2} (c_{\mathbf{r},A} + i c_{\mathbf{r},B}) \quad (18)$$

then its Fourier transform would be

$$\gamma_k = \frac{1}{\sqrt{2N}} \sum_{\mathbf{r}} \gamma_{\mathbf{r}} e^{-ik \cdot \mathbf{r}} = \frac{1}{2} (d_{k,A} + i d_{k,B})$$

In this way, the operator d can be expressed in terms of γ fermion,

$$\begin{cases} d_{k,A} = \gamma_{-\mathbf{k}}^\dagger + \gamma_k \\ d_{k,B} = i (\gamma_{-\mathbf{k}}^\dagger - \gamma_k) \end{cases}$$

Using the γ operators to rewrite the Hamiltonian gives

$$\hat{H} = \sum_{\mathbf{k}} \left[R_{\mathbf{k}} \cdot (\gamma_{\mathbf{k}}^\dagger \gamma_{\mathbf{k}} - \gamma_{-\mathbf{k}} \gamma_{-\mathbf{k}}^\dagger) - i I_{\mathbf{k}} \cdot (\gamma_k^\dagger \gamma_{-\mathbf{k}}^\dagger - \gamma_{-\mathbf{k}} \gamma_k) \right] = \sum_{\mathbf{k}} \phi_k^\dagger \mathbf{H}(k) \phi_k$$

where $R_{\mathbf{k}} \equiv \text{Re} \{s_{\mathbf{k}}\}$ and $I_{\mathbf{k}} \equiv \text{Im} \{s_{\mathbf{k}}\}$, and

$$\mathbf{H}(k) \equiv \begin{pmatrix} R_{\mathbf{k}} & -i I_{\mathbf{k}} \\ i I_{\mathbf{k}} & -R_{\mathbf{k}} \end{pmatrix}, \quad \phi_k \equiv \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma_{-\mathbf{k}}^\dagger \end{pmatrix}$$

Since

$$\mathbf{H}(k)^2 = (R_{\mathbf{k}}^2 + I_{\mathbf{k}}^2) \mathbb{1} = |s_{\mathbf{k}}|^2 \mathbb{1}$$

we can see that the eigenvalues are $\epsilon_k = \pm |s_{\mathbf{k}}|$, which is consistent with Kitaev's result (equation (32) in Ref.[?]). To diagonalize the matrix $\mathbf{H}$ we have to apply the Bogoliubov transformation,

$$U = \frac{1}{\sqrt{2|s_{\mathbf{k}}|}} \begin{pmatrix} \text{sgn}(I_k)\sqrt{|s_{\mathbf{k}}| + R_{\mathbf{k}}} & -i\sqrt{|s_{\mathbf{k}}| - R_{\mathbf{k}}} \\ i\sqrt{|s_{\mathbf{k}}| - R_{\mathbf{k}}} & -\text{sgn}(I_k)\sqrt{|s_{\mathbf{k}}| + R_{\mathbf{k}}} \end{pmatrix} = \begin{pmatrix} \text{sgn}(I_k)u_k & -iv_k \\ iv_k & -\text{sgn}(I_k)u_k \end{pmatrix}$$

$$= v_k\sigma^y + \text{sgn}(I_k)u_k\sigma^z$$

such that²

$$U\mathbf{H}U^\dagger = \begin{pmatrix} |s_{\mathbf{k}}| & 0 \\ 0 & -|s_{\mathbf{k}}| \end{pmatrix}$$

Therefore, the excitation that diagonalize the Hamiltonian is

$$\begin{pmatrix} a_k \\ a_{-k}^\dagger \end{pmatrix} = U \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma_{-\mathbf{k}}^\dagger \end{pmatrix} = \begin{pmatrix} \text{sgn}(I_k)u_k\gamma_{\mathbf{k}} - iv_k\gamma_{-\mathbf{k}}^\dagger \\ iv_k\gamma_{\mathbf{k}} - \text{sgn}(I_k)u_k\gamma_{-\mathbf{k}}^\dagger \end{pmatrix} \quad (19)$$

As a result, the Kitaev ground state is the vacuum state of the Bogoliubov fermion, i.e., $a_k|M\rangle = 0$, which means $|M\rangle$ can be written as a BCS ground state just as we met in the BCS theory. The ground state can be written as the coherent state³ of the γ -fermion pair,

$$\begin{aligned} |M\rangle &= A[g] \exp\left(\sum_{\mathbf{k}} g_{\mathbf{k}} \gamma_{\mathbf{k}}^\dagger \gamma_{-\mathbf{k}}^\dagger\right) |\text{vac}_\gamma\rangle \\ &= A[g] \exp\left(\sum_{\mathbf{r}\mathbf{r}'} g_{\mathbf{r}\mathbf{r}'} \gamma_{\mathbf{r}}^\dagger \gamma_{\mathbf{r}'}^\dagger\right) |\text{vac}_\gamma\rangle \\ &= A[g] \left[\mathbb{1} + \sum_{\mathbf{r}\mathbf{r}'} g_{\mathbf{r}\mathbf{r}'} \gamma_{\mathbf{r}}^\dagger \gamma_{\mathbf{r}'}^\dagger + \frac{1}{2!} \sum_{\mathbf{r}\mathbf{r}'\kappa\lambda} g_{\mathbf{r}\mathbf{r}'} g_{\kappa\lambda} \gamma_{\mathbf{r}}^\dagger \gamma_{\mathbf{r}'}^\dagger \gamma_{\kappa}^\dagger \gamma_{\lambda}^\dagger + \dots \right] |\text{vac}_\gamma\rangle \end{aligned}$$

where $|\text{vac}_\gamma\rangle$ is the vacuum state of the γ -fermions, and $g_{\mathbf{r}\mathbf{r}'} = \frac{1}{2N} \sum_{\mathbf{k}} g_{\mathbf{k}} e^{-ik\cdot(\mathbf{r}-\mathbf{r}')}.$

A.1 With External Magnetic Field

For simplicity, we focus only on the isotropic limit [12]. In this situation, applying an external magnetic field $\mathbf{h} = (h_x, h_y, h_z)$, which is equivalent to adding a Zeeman perturbation $H' = -\sum_i \mathbf{h} \cdot \boldsymbol{\sigma}$, makes the Kitaev model equivalent to the Haldane honeycomb model [4] up to the third-order perturbation. The Hamiltonian under this approximation in the Majorana fermion representation

²Those parameters in the matrix U have the following properties: (i) $u_k = u_{-k}$, (ii) $v_k = v_{-k}$, and (iii) $\text{sgn}(I_k) = -\text{sgn}(I_{-k})$

³The coefficient g_k can be represented in terms of u_k and v_k . As I have mentioned, the BCS-like ground state is expected to be the vacuum state of a_k fermions for all possible k , thus, we have

$$\left(\text{sgn}(I_k)u_k\gamma_{\mathbf{k}} - iv_k\gamma_{-\mathbf{k}}^\dagger\right) \prod_k \left(\mathbb{1} + g_{\mathbf{k}}\gamma_{\mathbf{k}}^\dagger \gamma_{-\mathbf{k}}^\dagger\right) |\text{vac}_\gamma\rangle = 0$$

Based on this, we obtain that

$$\text{sgn}(I_k)u_k g_k - iv_k = 0$$

therefore

$$g_k = \text{sgn}(I_k) i \frac{v_k}{u_k} = \text{sgn}(I_k) i \sqrt{\frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{|s_{\mathbf{k}}| + R_{\mathbf{k}}}} = i \frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{I_k} \quad (20)$$

reads

$$\begin{aligned}
H &= -\frac{1}{2}J \sum_{\mathbf{r}} (i c_{\mathbf{r}A} c_{\mathbf{r}+\hat{\mathbf{n}}_1, B} + i c_{\mathbf{r}A} c_{\mathbf{r}+\hat{\mathbf{n}}_2, B} + i c_{\mathbf{r}A} c_{\mathbf{r}B} + \text{H.c.}) \\
&\quad - \frac{1}{2}\kappa \sum_{\mathbf{r}} \sum_{\mu=A, B} (c_{\mathbf{r}\mu} c_{\mathbf{r}+\hat{\mathbf{n}}_1, \mu} + c_{\mathbf{r}\mu} c_{\mathbf{r}+\hat{\mathbf{n}}_2, \mu} + c_{\mathbf{r}\mu} c_{\mathbf{r}+\hat{\mathbf{n}}_1-\hat{\mathbf{n}}_2, \mu} + \text{H.c.}) \\
H &= \frac{1}{2} \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^{\dagger} \begin{pmatrix} \Delta_{\mathbf{k}} & i s_{\mathbf{k}} \\ -i s_{\mathbf{k}}^* & -\Delta_{\mathbf{k}} \end{pmatrix} \Psi_{\mathbf{k}} \\
&\quad \Delta_{\mathbf{k}} =
\end{aligned}$$

B Sum Rules for Correlation Functions

$$\begin{aligned}
\rho &= \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} \\
\mathcal{P}^{(p)} &= \left(\frac{\mathbb{1} + \tilde{W}_v}{2} \right) \left(\frac{\mathbb{1} + \tilde{W}_h}{2} \right) \prod_p \left(\frac{\mathbb{1} + W_p}{2} \right) \\
\rho^2 &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \varphi(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\
&= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \frac{1}{2} [\varphi(\mathfrak{D}, \mathfrak{D}') + \varphi(\mathfrak{D}', \mathfrak{D})] \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\
&= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\
&= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}^c]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}^c \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} \vartheta(\mathfrak{D}^c \ominus \mathfrak{D}', \mathfrak{D}') \Sigma_{\mathfrak{D}^c} \\
&= \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \left(\frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} \eta_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}') \right) \Sigma_{\mathfrak{D}} \\
&\quad \frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}']} \vartheta(\mathfrak{D}, \mathfrak{D}') \mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} = \mathfrak{C}_{\mathfrak{D}} \tag{21} \\
&\quad \frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}}^2 = 1 \tag{22}
\end{aligned}$$

notice that $\eta_{\mathfrak{D}} = 1$ if $[\mathfrak{D}] \in \mathbf{G}$

$$\vartheta(\mathfrak{D}, \mathfrak{D}') = \frac{1}{2} [1 + \eta_{\mathfrak{D}} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D} \ominus \mathfrak{D}'}] \varphi(\mathfrak{D}, \mathfrak{D}') = \vartheta(\mathfrak{D}', \mathfrak{D}) = 0 \text{ or } \pm 1$$

$$\varphi(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \eta_{\mathfrak{D}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \varphi(\mathfrak{D}', \mathfrak{D}) = \eta_{\mathfrak{D}'} \varphi(\mathfrak{D}, \mathfrak{D}')$$

$$\vartheta(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \frac{1}{2} [1 + \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D}}] \varphi(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \eta_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}')$$

Lemma B.1. If $[\mathfrak{D}] \in \mathbf{G}$, then the following identity holds

$$\sum_{[\mathfrak{D}'] \in \mathbf{G}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} = \delta_{[\mathfrak{D}], [\emptyset]}$$

Proof. By definition, the factor $\eta_{\mathfrak{D}}$ can be expressed as $\eta_{\mathfrak{D}} = (-1)^{\Delta(\mathfrak{D})/2}$, where $\Delta(\mathfrak{D}) \equiv n_a(\mathfrak{D}) - n_b(\mathfrak{D})$.

$$\begin{aligned} \Delta(\mathfrak{D} \ominus \mathfrak{D}') &= \Delta(\mathfrak{D}) + \Delta(\mathfrak{D}') - n_a(\mathfrak{D} \cap \mathfrak{D}') + n_b(\mathfrak{D} \cap \mathfrak{D}') \\ \eta_{\mathfrak{D} \ominus \mathfrak{D}'} &= (-1)^{\Delta(\mathfrak{D} \ominus \mathfrak{D}')} = \eta_{\mathfrak{D}} \eta_{\mathfrak{D}'} (-1)^{n_a(\mathfrak{D} \cap \mathfrak{D}')} (-1)^{n_b(\mathfrak{D} \cap \mathfrak{D}')} = (-1)^{n_a(\mathfrak{D} \cap \mathfrak{D}')} (-1)^{n_b(\mathfrak{D} \cap \mathfrak{D}')} \\ \sum_{[\mathfrak{D}'] \in \mathbf{G}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} &= \sum_{[\mathfrak{D}'] \in \mathbf{G}} (-1)^{n_a(\mathfrak{D} \cap \mathfrak{D}')} (-1)^{n_b(\mathfrak{D} \cap \mathfrak{D}')} \end{aligned}$$

■

Lemma B.2. If $[\mathfrak{D}] \in \mathbf{G}$, then the following identity holds

$$\sum_{[\mathfrak{D}'] \in \mathbf{G}} \Sigma_{\mathfrak{D}'} \Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \delta_{[\mathfrak{D}], [\emptyset]} \mathbb{1}$$

Proof. According to the identity $\varphi(\mathfrak{D}, \mathfrak{D}') = \eta_{\mathfrak{D}} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \varphi(\mathfrak{D}', \mathfrak{D})$, we have the commutation relation

$$\Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \Sigma_{\mathfrak{D}'} \Sigma_{\mathfrak{D}}$$

In the above identity, we have already considered $[\mathfrak{D}], [\mathfrak{D}'] \in \mathbf{G}$ in our case. With this relation, we could rewrite the summation in the equation we wanted to prove

$$\sum_{[\mathfrak{D}'] \in \mathbf{G}} \Sigma_{\mathfrak{D}'} \Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \sum_{[\mathfrak{D}'] \in \mathbf{G}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \Sigma_{\mathfrak{D}}$$

■

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